

DAD FieldWorks

A Reusable Evidence Contract Architecture for Validation Aware Computational Electromagnetics Software

Technical Whitepaper / Public Specification Draft

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This document describes a research and development architecture for validation aware computational electromagnetics software. It is not a certified validation report, not a product release, not an external solver equivalence claim, and not a production readiness statement. The concepts, terminology, structure, and architectural presentation are provided for technical discussion and documentation purposes. No part of this document may be reproduced, redistributed, or used as an official validation statement without written permission from the copyright holder.

Abstract

Computational electromagnetics software often produces highly precise numerical values while leaving the evidential status of those values implicit. This can lead to premature or unsupported claims regarding solver correctness, validation, external equivalence, or production readiness. This paper proposes a reusable evidence contract architecture for validation aware computational electromagnetics software. The architecture separates numerical computation from evidence, gate evaluation, decision classification, failure classification, reference comparison, risk tracking, and claim boundaries. It is motivated by a PEC cavity investigation within the DAD FieldWorks project, where analytical cavity theory, scalar Helmholtz eigenvalue prototypes, numerical dispersion diagnostics, time domain ringdown evidence, and feature tracking ambiguity produced distinct but related evidence paths. A C++ evidence contract foundation is introduced as a future C++ core interface. The proposed approach does not itself validate any solver. Instead, it defines a disciplined structure by which future eigenmode, FDTD, FEM, MoM, and external comparison workflows may record evidence and prevent unsupported claims.

Keywords: computational electromagnetics, CEM, VVUQ, validation, verification, uncertainty quantification, evidence contracts, FDTD, eigenmode solver, Helmholtz, Method of Moments, finite element method, external solver comparison, C++ software architecture

1 Introduction

Computational electromagnetics, hereafter CEM, is frequently concerned with numerical results that appear precise but whose evidential meaning depends on many hidden conditions. A resonance frequency, field distribution, impedance value, current profile, or scattering parameter is not automatically a validated engineering result. It is a numerical result produced under a specific model, grid, boundary condition, material assumption, numerical method, and post processing chain.

In practical engineering software this distinction is often blurred. A solver may output a value without making clear whether the result is analytically checked, compared against measurement, compared against an external code, restricted to a canonical problem, or merely generated by an internal prototype.

The central thesis of this paper is that CEM software should not only return numerical values. It should return structured evidence. A solver result should carry its provenance, assumptions, comparisons, gates, risks, limitations, and allowed claims.

The proposed architecture is called an evidence contract. It is designed to support validation aware development without prematurely claiming validation.

2 Motivation

2.1 The Problem of Unqualified Numerical Results

A numerical solver output may be technically correct within its model and still be misinterpreted. This problem occurs when one of the following questions remains unanswered:

- Which mathematical model produced the result?
- Which source or equation supports that model?
- Which assumptions were active?
- Which boundary conditions were applied?
- Which grid, mesh, port, source, probe, or window was used?
- Was the result compared to an analytical reference?
- Was it compared to a canonical benchmark?
- Was it compared to an external solver?
- Was it compared to measurement?
- Is the result allowed to support a validation claim?
- Is the result allowed to support production use?

If these questions are not answered, the numerical value stands alone. It may be useful, but its engineering meaning is incomplete.

2.2 Validation Aware Software

Validation aware software does not assume that every successful computation is a validated result. It distinguishes:

- internal numerical execution,
- analytical comparison,
- canonical comparison,
- external code comparison,
- measurement comparison,
- production readiness.

The evidence contract architecture proposed here is intended to make this distinction machine readable.

3 Relationship to CEM Validation Practice

Computational electromagnetics validation is not a simple assertion of correctness. It is usually based on structured comparison against one or more trusted evidence sources, such as analytical methods, canonical solutions, measurements, or alternative computational codes.

This paper does not claim compliance with any validation standard and does not claim that DAD FieldWorks is externally validated. Instead, it adopts the methodological principle that comparison type, evidence source, and claim scope must be explicit.

Therefore, the proposed evidence architecture treats the following as distinct evidence channels:

- analytical reference,
- canonical method,
- external solver result,
- measured data,
- internal numerical diagnostic,
- bounded prototype execution.

4 Core Concept: Evidence Contract

4.1 Definition

An evidence contract is a structured result object that couples numerical output with its evidential state. In abstract form:

$$\mathcal{R}_{\text{evidence}} = (M, S, E, G, C, B, R, F)$$

where:

- M denotes solver and case metadata,
- S denotes result contract summary,

- E denotes evidence rows,
- G denotes gate rows,
- C denotes comparison rows,
- B denotes claim boundary rows,
- R denotes risk rows,
- F denotes required future work rows.

The key point is that a solver result is not reduced to a scalar number. It becomes a structured evidence bearing object.

4.2 Conservative Claim Defaults

The default claim state is conservative:

ExternallyValidatedQ = False

ProductionAllowedQ = False.

These flags should not become true by numerical success alone. They require explicit supporting evidence.

5 Evidence Row Model

An evidence row is a machine readable statement about the origin, method, assumption, observation, or limitation of a result.

A generic evidence row may contain:

- EvidenceId,
- Category,
- Name,
- Value,
- Units,
- Source,
- Notes.

Evidence rows are intentionally general. They can represent model assumptions, grid metadata, source verification, solver settings, boundary conditions, residuals, or interpretation notes.

6 Gate Row Model

A gate row is a structured check. It answers whether a required condition is satisfied.

A minimal gate row contains:

- Gate,
- PassQ,
- Value,
- Notes.

For example:

Gate = ReferenceAvailableGate

PassQ = True

or

Gate = ProductionAllowedDefaultFalseGate

PassQ = True.

A gate row should be explicit. Hidden assumptions should not count as passed gates.

7 Decision and Failure Classes

A decision class summarizes the state of a result. A failure class summarizes why a result is blocked, limited, or not yet accepted.

Example decision classes include:

- Executed,
- Pass,
- Warning,
- Fail,
- Partial,
- Inconclusive,
- Blocked.

Example failure classes include:

- MissingReference,
- MissingMetadata,
- GeometryMismatch,
- BoundaryMismatch,
- SolverNotExecuted,
- AmbiguousEvidence,
- ClaimBoundaryViolation,
- ResidualTooLarge,
- ExternalComparisonMissing.

The purpose is not bureaucracy. The purpose is to prevent silent ambiguity.

8 Reference Comparison

Reference comparison is a first class part of the evidence contract.

For a scalar quantity such as frequency:

$$\Delta f = |f_{\text{num}} - f_{\text{ref}}|$$

and

$$\epsilon = \frac{|f_{\text{num}} - f_{\text{ref}}|}{f_{\text{ref}}}.$$

A reference comparison row should record:

- reference type,
- reference source,
- reference value,
- observed value,
- absolute error,
- relative error,
- threshold policy,
- comparison status.

The comparison type must be explicit. Analytical comparison, external solver comparison, and measurement comparison are not interchangeable.

9 Claim Boundary Model

A claim boundary row states what the result does not claim.

Typical claim boundary rows include:

- no external validation claim,
- no production readiness claim,
- no solver equivalence claim,
- no measurement agreement claim,
- no full Maxwell solver claim for scalar prototypes,
- no FDTD validation claim from an eigenmode prototype.

This is essential because many numerical results are useful but limited.

10 Reusable VVUQ Patterns

The PEC cavity investigation led to the extraction of reusable VVUQ evidence patterns.

10.1 SourceVerificationPattern

Purpose:

- verify the source of an equation or model,
- document bibliographic metadata,
- record equation references,
- document assumptions and validity range,
- avoid copying copyrighted text or code.

10.2 ReferenceArtifactPattern

Purpose:

- store reference values in machine readable form,
- record units,
- record geometry and boundary conditions,
- keep reference values separate from solver outputs.

10.3 AnalyticalReferenceComparisonPattern

Purpose:

- compare numerical results to analytical references,
- compute absolute error,
- compute relative error,
- apply threshold policies,
- prevent premature validation claims.

10.4 ExternalSolverComparisonPattern

Purpose:

- compare internal results against external solvers,
- ensure geometry equivalence,
- ensure material equivalence,
- ensure port equivalence,
- compare compatible frequency grids,
- avoid unsupported equivalence claims.

10.5 ExecutionArtifactPattern

Purpose:

- record bounded solver execution,
- store scalar summaries and machine readable rows,
- avoid unnecessary full volume storage,
- preserve reproducibility.

10.6 EvidenceReviewPattern

Purpose:

- inspect already generated records,
- avoid recomputation,
- confirm artifacts are readable,
- summarize what is known and what remains uncertain.

10.7 GridMetadataReadinessPattern

Purpose:

- verify grid dimensions,
- verify cell sizes,
- verify coordinate policies,
- verify boundary masks,
- verify field component layout.

10.8 NumericalDispersionDiagnosticPattern

Purpose:

- diagnose numerical dispersion,
- distinguish continuous reference error from discrete grid behavior,
- reduce false solver defect claims.

10.9 AmbiguityAuditPattern

Purpose:

- identify ambiguous evidence,
- preserve competing explanations,
- prevent silent candidate promotion.

10.10 PeakOrFeatureTrackingPattern

Purpose:

- track peaks, modes, resonances, or features,
- record rank, amplitude, component, probe, window, and provenance,
- avoid raw maximum only decisions.

10.11 ObservableDependencePattern

Purpose:

- detect dependency on observable type,
- detect component dependency,
- detect probe dependency,
- detect window dependency.

10.12 ClaimBoundaryPattern

Purpose:

- state what is not claimed,
- keep validation and production flags conservative,
- prevent accidental claim expansion.

11 Application to Solver Families

11.1 Eigenmode Solvers

For an eigenmode solver, evidence contracts should include:

- eigenvalue problem formulation,
- boundary conditions,
- matrix size,
- eigenvalue,
- eigenfrequency,
- mode classification,
- residual norm,
- spurious mode checks,
- reference comparison.

11.2 FDTD Solvers

For FDTD solvers, evidence contracts should include:

- grid rows,
- CFL rows,
- source rows,
- probe rows,
- boundary rows,
- energy rows,
- dispersion rows,
- time window rows,
- peak or feature tracking rows,
- claim boundary rows.

11.3 FEM Helmholtz Solvers

For FEM Helmholtz solvers, evidence contracts should include:

- weak form records,
- mesh metadata,
- boundary condition rows,
- material rows,
- residual rows,
- convergence rows,
- analytical or manufactured solution comparison.

11.4 MoM Antenna Solvers

For Method of Moments antenna solvers, evidence contracts should include:

- geometry rows,

- segmentation rows,
- basis and testing function rows,
- feed model rows,
- impedance rows,
- current distribution rows,
- comparison to analytical, institutional, or external solver references where available.

11.5 External Solver Workflows

For external solver workflows, evidence contracts should include:

- same geometry rows,
- material equivalence rows,
- mesh equivalence rows,
- port equivalence rows,
- frequency grid rows,
- result import rows,
- comparison metric rows,
- no equivalence claim unless explicitly supported.

12 C++ Evidence Contract Foundation

The reusable VVUQ methodology has been mapped into C++ header only infrastructure.

The implemented core types are:

- EvidenceRow,
- GateRow,
- DecisionClass,
- FailureClass,
- ClaimBoundary,
- ReferenceComparisonRow,
- RiskRow,
- RequiredFutureWorkRow,
- SolverMetadata,
- ValidationArtifactMetadata,
- ResultContractSummary,
- SolverEvidenceResult.

The top level object is:

```
SolverEvidenceResult.
```

It is intended to carry:

- metadata,
- summary,

- evidence rows,
- gate rows,
- reference comparison rows,
- risk rows,
- claim boundary rows,
- required future work rows.

The conservative defaults are:

```
ExternallyValidatedQ = false
```

```
ProductionAllowedQ = false.
```

This brings the evidence discipline into the future C++ implementation path.

13 C++ Demo Result Wrapper

A narrow C++ demo wrapper was added:

```
make_scalar_helmholtz_eigenmode_demo_result()
```

in namespace:

```
dadfw::vvuq::examples.
```

It returns a `SolverEvidenceResult` with metadata:

- SolverFamily: Eigenmode,
- SolverName: ScalarHelmholtzEigenmodeDemo,
- Method: ResultContractDemoOnly,
- CaseId: PEC_CAVITY_SCALAR_HELMHOLTZ_DEMO.

The demo adds:

- evidence rows declaring demo only status,
- gates confirming no solver numerics were executed,
- reference readiness rows without numerical comparison,
- claim boundary rows,
- risk rows,
- required future work rows.

It does not solve eigenvalues and does not compute eigenfrequencies. It exists only to show how future C++ solvers should package their results.

14 Case Study: PEC Cavity

14.1 Analytical Reference

For a rectangular PEC cavity the analytical frequency is

$$f_{mnp} = \frac{c_0}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2}.$$

For the investigated cavity:

$$a = 0.08 \text{ m}, \quad b = 0.084 \text{ m}, \quad d = 0.084 \text{ m},$$

$$m = n = p = 1.$$

With

$$c_0 = 299792458 \text{ m/s},$$

the reference frequency is

$$f_{\text{ref}} = 3143165987.503105 \text{ Hz.}$$

or

$$f_{\text{ref}} = 3.143165987503105 \text{ GHz.}$$

14.2 Route A Scalar Helmholtz Prototype

The scalar Helmholtz prototype solves

$$-\nabla^2 u = \lambda u$$

with homogeneous Dirichlet boundary conditions.

The prototype uses

$$k = \sqrt{\lambda}$$

and

$$f = \frac{c_0 k}{2\pi}.$$

The first prototype used an interior grid:

$$\{9, 8, 8\}.$$

The operator size was

$$576 \times 576$$

with

$$3616$$

nonzero entries.

The selected eigenvalue was

$$\lambda = 4298.700258260234.$$

The selected eigenfrequency was

$$f_{\text{SH}} = 3.128307359057517 \text{ GHz.}$$

The relative error against the analytical reference was

$$\epsilon_{\text{SH}} = 0.0047272808705185925.$$

The residual norm was

$$7.0051762275510885 \times 10^{-15}.$$

This is an internal prototype result, not external validation.

14.3 Scalar Helmholtz Grid Refinement

A bounded scalar Helmholtz grid refinement study was executed. It showed monotonic convergence toward the analytical reference.

The optional level

$$\{36, 32, 32\}$$

was blocked because it exceeded the bounded execution limit.

The refinement result showed:

Relative error decreases = true

Monotonic convergence = true.

14.4 Same Grid Scalar Eigenmode Diagnostic

A bounded same grid scalar eigenmode diagnostic was executed to examine how the scalar Dirichlet eigenproblem behaves when existing DAD field grid metadata are used.

Three mapping policies were evaluated:

- Policy 1: grid dimensions interpreted as scalar interior unknown counts.
- Policy 2: grid dimensions interpreted as cell counts with a Dirichlet interior convention.
- Policy 3: cell sizes used directly to compute effective scalar domain lengths.

The same grid result was classified as inconclusive because all policies solved cleanly, but the spread between mapping policies remained material. Policy 2 and Policy 3 are near the analytical reference and the Route A scalar trend, while Policy 1 is far enough to keep grid mapping ambiguity active.

This shows that the numerical eigenvalue solves are clean, but the interpretation of grid metadata and boundary conventions must be documented more rigorously before any same grid scalar policy can be selected.

Table 1: Scalar Helmholtz grid refinement against analytical PEC cavity reference

Level	Grid	Frequency / GHz	Relative Error	Residual Norm
0	{9, 8, 8}	3.128307359058	0.0047272809	2.63×10^{-15}
1	{18, 16, 16}	3.139012096545	0.0013215627	1.29×10^{-14}
2	{27, 24, 24}	3.141247276062	0.0006104391	1.68×10^{-14}

Table 2: Same grid scalar eigenmode diagnostic

Policy	Matrix Size	Frequency / GHz	Relative Error	Residual Norm
Policy 1	504×504	2.778104015200	0.1161446687	1.83×10^{-15}
Policy 2	336×336	3.122528486251	0.0065658325	1.14×10^{-15}
Policy 3	504×504	3.126946915965	0.0051601066	2.37×10^{-15}

14.5 Comparison of Evidence Paths

The scalar Helmholtz eigenvalue path is much closer to the analytical reference than the observed FDTD ring-down families. More importantly, it exhibits monotonic convergence under grid refinement. The same grid scalar diagnostic shows that plausible scalar mappings remain near the analytical reference, but also that grid convention ambiguity is material and must not be hidden.

15 Advantages of the Proposed Architecture

15.1 Traceable Engineering Results

A DAD FieldWorks result is intended to answer not only: *What is the number?*

It should also answer: *Why should this number be trusted, and within which limits?*

15.2 Modular Solver Architecture

The planned architecture separates:

- RF model kernels,
- solver kernels,
- validation wrappers,
- reference comparisons,
- external solver workflows,
- documentation artifacts,
- C++ result contracts.

15.3 Better Failure Visibility

The framework is designed to distinguish:

- model error,
- grid error,
- source error,
- probe error,

- windowing error,
- feature selection error,
- solver implementation error.

15.4 Scalable Evidence Model

The same evidence model can later support:

- eigenmode solvers,
- FDTD solvers,
- FEM Helmholtz solvers,
- MoM antenna solvers,
- external comparison workflows,
- C++ production kernels.

16 Limitations

The current state remains research and development. The following claims are not made:

- no external validation,
- no production readiness,
- no commercial solver equivalence,
- no certified external solver agreement,
- no full vector Maxwell eigenmode solver,
- no native Yee curl curl solver,
- no externally validated FDTD solver.

17 Roadmap

17.1 Near Term

- review the same grid scalar interpretation,
- document grid convention and boundary mapping rules,
- continue local C++ compilation and contract testing,
- keep validation and production flags conservative.

Table 3: PEC cavity frequency comparison

Method or family	Frequency / GHz	Relative error to analytical reference
Analytical reference	3.143 165 987 503	0.000 000 000
Route A scalar Helmholtz Level 0	3.128 307 359 058	0.004 727 281
Route A scalar Helmholtz Level 1	3.139 012 096 545	0.001 321 563
Route A scalar Helmholtz Level 2	3.141 247 276 062	0.000 610 439
Same grid scalar Policy 2	3.122 528 486 251	0.006 565 833
Same grid scalar Policy 3	3.126 946 915 965	0.005 160 107
Numerical dispersion diagnostic	3.126 477 519 038	0.005 309 445
FDTD EzHxFamily	2.950 641 806 786	0.061 251 675
FDTD ExFamily	3.631 559 146 814	0.155 382 554

17.2 Mid Term

- plan native Yee curl curl metadata requirements,
- document native field component offsets and boundary masks,
- plan a native Yee curl curl eigenmode prototype,
- define FDTD reference cases independent of PEC feature tracking.

17.3 Long Term

- create a full eigenmode solver chain,
- establish independent FDTD validation cases,
- build external comparison pipelines,
- migrate mature research kernels into C++,
- keep evidence contracts mandatory for all solver families.

18 Conclusion

This paper proposed a reusable evidence contract architecture for validation aware computational electromagnetics software. The architecture separates computation from evidence, comparison, risk, decision classification, failure classification, and claim boundaries.

The PEC cavity investigation demonstrated why this separation matters. A direct analytical eigenmode reference, a scalar Helmholtz eigenvalue prototype, a numerical dispersion diagnostic, a same grid scalar diagnostic, and a source driven FDTD ringdown workflow are not the same evidence path. Treating them as identical would lead to false conclusions.

The reusable VVUQ evidence patterns and the C++ `SolverEvidenceResult` foundation turn this lesson into an architectural principle. The Route A scalar Helmholtz prototype and its monotonic grid refinement show a clean numerical eigenvalue path, separated from time domain feature tracking. The same grid scalar diagnostic shows that grid mapping conventions matter and must be documented before stronger claims can be made.

The proposed architecture does not claim validation by itself. It defines the structure required so that future validation claims can be made only when evidence actually supports them.

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